

CREATOR PLANET

Build • Connect • Create

Phase 1 – Game Design Document (GDD)

Chosen Theme	Creator Planet
Primary Category	World-Building Game
Target Age / Grade	10–16 years / Grades 5–10
Player Role	Planet Creator
Technology	Unity + C#
Platform	Windows PC
Prototype Scope	One stylized planet, 2–3 areas, limited reusable objects, 3 challenges

1. Game Concept

Creator Planet is a 3D creative problem-solving game where the player receives a small, partially empty planet and transforms it into a functional world. The player directly places, moves, rotates, removes and connects environmental objects. Each construction choice affects whether the planet's inhabitants can complete a challenge.

2. Discovery World / Theme Fit

The game uses the **Creator Planet** theme by making the player an active creator of a living 3D world. Instead of only exploring a pre-built environment, the player changes the environment itself. The planet becomes the player's creative workspace and the result of their decisions.

3. Target Age / Grade and Player Role

Target: Ages 10–16 / Grades 5–10.

Player role: The player is a **Planet Creator** responsible for designing a small planet, connecting important locations and making the environment usable for its inhabitants.

4. Problem / Opportunity

Many learners can benefit from interactive activities that combine creativity with practical problem-solving. Creator Planet provides an opportunity to practice **planning, spatial reasoning, creativity, sequencing and problem-solving** through direct 3D world construction.

5. Player Objective

Build and configure a functional planet by placing, modifying and connecting 3D world elements so that the planet's inhabitants can complete three construction challenges.

6. One-Sentence Core Gameplay Loop

Explore → Build → Modify → Connect → Test → Solve → Unlock the next challenge.

7. World-Building Category Evidence

The game directly demonstrates the **World-Building Game** category: the player can **place, move, rotate, remove and meaningfully configure 3D world elements** such as roads, bridges, buildings and power stations. The player must connect these elements to solve challenges, so world modification is the core gameplay rather than decoration.

8. Main Controls and Player Actions

- **WASD:** Move around the planet.
- **Mouse:** Look / control the camera.
- **Left click:** Select and place an object.
- **Drag:** Move a selected object.
- **R:** Rotate an object.
- **Delete:** Remove an object.
- **Space:** Test the created world.

9. 3D Locations

Creator Island: Main construction area where the player builds the settlement.

Energy Valley: Area containing the power station and buildings that need a functional connection.

Split Islands: Two small land areas separated by water; the player must construct a bridge/path to connect them.

10. Buildable / Configurable Objects

The prototype will use a limited reusable set of stylized 3D objects: **houses, roads, bridges, power stations, trees, water sources, resource stations and gates**. Objects can be repositioned and rotated, and selected objects can be removed. Roads, bridges and power-related elements can be configured to satisfy challenge requirements.

11. Funobotz Character Integration

Funobotz acts as the player's interactive Planet Helper Robot. It introduces challenges, explains objectives, provides hints when a construction is incorrect, and celebrates successful solutions. This matters to gameplay because the robot is the main feedback and guidance system rather than a decorative character.

12. Progress and Win Condition

The prototype contains three challenges:

Challenge 1: Build a small settlement.

Challenge 2: Connect the settlement to a power station.

Challenge 3: Connect two islands using a bridge/path.

Completing all three challenges triggers a **Planet Complete** result and a Creator Score / completion badge.

13. Feedback and Failure Condition

The game provides immediate visual and Funobotz feedback. A challenge fails when required objects are missing, incorrectly connected, or when the destination cannot be reached. Instead of a harsh game-over screen, Funobotz gives a contextual hint such as: *“The bridge is not connected to both islands. Try repositioning it.”*

14. Replayability

Players can rebuild the same challenge using different layouts and routes. They can move or replace objects after completion, experiment with alternative solutions, and replay challenges to improve their Creator Score. This encourages experimentation rather than one fixed construction sequence.

15. Technology / Platform

Game Engine: Unity

Programming: C#

3D Assets: Blender and/or Unity Asset Store assets

Development Environment: Visual Studio / Visual Studio Code

Target Platform: Windows PC

16. AI-Tool Declaration

AI tools may be used during development for brainstorming, code assistance, debugging guidance, writing support and placeholder asset ideas. The development team will perform the final implementation, integration, testing and validation of all game features.

17. Realistic 24-Hour Scope

The prototype is intentionally limited to one stylized planet, 2–3 small areas, a reusable set of 3D objects, basic player movement/camera, object placement and transformation, three challenges, Funobotz guidance, win/failure feedback and simple UI. Advanced multiplayer, procedural planet generation, large open worlds, complex NPC simulation and VR are outside the 24-hour scope.

18. 24-Hour Development Plan

Time	Development Task
0–2 hrs	Unity project setup, basic planet and scene
2–5 hrs	Player movement and camera controls
5–9 hrs	Object selection and placement system
9–12 hrs	Move, rotate and delete interactions
12–15 hrs	Implement three building challenges
15–17 hrs	Funobotz guidance and feedback
17–20 hrs	Win/failure logic and UI
20–22 hrs	Sound, visual polish and usability fixes
22–24 hrs	Full testing, bug fixing and final build

19. Phase 1 Rubric Evidence Summary

Discovery World Fit: The Creator Planet theme is expressed through active 3D planet creation and transformation.

Game Clarity: The player role, objective, controls, locations, gameplay loop, progression and win/failure conditions are explicitly defined.

Category Fit: Placing, modifying, connecting and configuring 3D environment elements is the central mechanic.

Originality / Problem Value: Creative construction is combined with spatial and practical problem-solving challenges.

24-Hour Feasibility: The prototype is constrained to one small planet, a limited asset set and three challenges.

Core Proof Statement: The player does not merely explore Creator Planet; the player changes the 3D world and must use those changes to solve meaningful challenges.

End of Phase 1 GDD